

****Glossary of Terms – Hotel Reservation System****

This glossary defines key terms, processes, and components used in the system specification.

Actors & Roles

1. **Guest**: A user who creates, modifies, or cancels a reservation.
 2. **Hotel Staff**: Personnel responsible for managing reservations, overbooking, and manual reviews.
 3. **Payment Gateway**: External service that processes financial transactions (e.g., credit card payments).
 4. **System**: Automated software managing reservation states, rules, and workflows.
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Reservation States

1. **Pending Confirmation**: Initial state after a reservation request; awaiting room availability verification.
 2. **Confirmed**: Reservation validated by the system; room secured.
 3. **Awaiting Payment**: Reservation pending guest payment within a deadline.
 4. **Payment Received**: Payment successfully processed; booking secured.
 5. **Checked-In**: Guest has arrived and started their stay.
 6. **In-Use**: Guest is actively occupying the room.
 7. **Extended**: Stay duration modified post-check-in (subject to availability).
 8. **Cancelled**: Reservation terminated before check-in (may incur fees).
 9. **No-Show**: Guest failed to check in by the deadline.
 10. **Checked-Out**: Stay completed; guest has left the hotel.
 11. **Under Review**: Reservation paused for manual investigation (e.g., fraud detection).
 12. **Refund Processed**: Refund issued per policy after cancellation.
 13. **Overbooked**: Reservation flagged due to excess bookings; requires resolution.
 14. **Maintenance Hold**: Room unavailable due to repairs; guest must be reassigned.
 15. **Expired**: Reservation invalidated due to inactivity (e.g., missed payment deadline).
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Processes & Policies

1. **Dynamic Pricing**: Automatic price adjustments based on demand, seasonality, or promotions.
 2. **Refund Policy**: Rules governing refund amounts based on cancellation timing (e.g., 100% refund >48h before check-in).
 3. **Loyalty Program**: Tiered rewards system granting priority during overbooking or upgrades.
 4. **Corporate/VIP Policies**: Custom rules for prioritized guests (e.g., extended payment deadlines, waived fees).
 5. **Overbooking Resolution**: Process to reallocate guests or provide compensation when rooms are oversold.
 6. **Prioritization Rules**: Hierarchy for resolving conflicts (e.g., loyalty members first).
 7. **Reassignment**: Moving a guest to a different room or hotel due to overbooking or maintenance.
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Technical Terms

1. **State Machine**: A model defining valid transitions between reservation states.
 2. **Idempotency**: Ensuring payment retries or duplicate requests do not cause unintended side effects.
 3. **Async Notifications**: Automated messages (email/SMS) sent via background processes.
 4. **Concurrency Control**: Managing simultaneous reservation updates to prevent conflicts.
 5. **Idempotent API**: An API design where repeated requests produce the same result (e.g., payment retries).
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Other Key Terms

1. **Check-In Deadline**: Timeframe by which a guest must arrive (e.g., 24 hours post-arrival date).
2. **Non-Refundable Booking**: Reservation type where cancellations forfeit payment.
3. **Compensation Package**: Offer to guests affected by overbooking (e.g., discounts, upgrades).
4. **Partner Hotel**: A third-party hotel used for guest reassignment during overbooking.
5. **Peak Season**: Periods of high demand with dynamic pricing surcharges.